EXPERIMENT 16

Implement a Sobel algorithm using OpenCV to filter the input image.

Aim

Implement a Sobel algorithm using OpenCV to filter the input image.

Algorithm

• Import OpenCV and read the image.

• Convert the image to grayscale.

• Calculate horizontal Sobel gradients.

• Calculate vertical Sobel gradients and combine them.

• Save the edge-filtered image.

Code

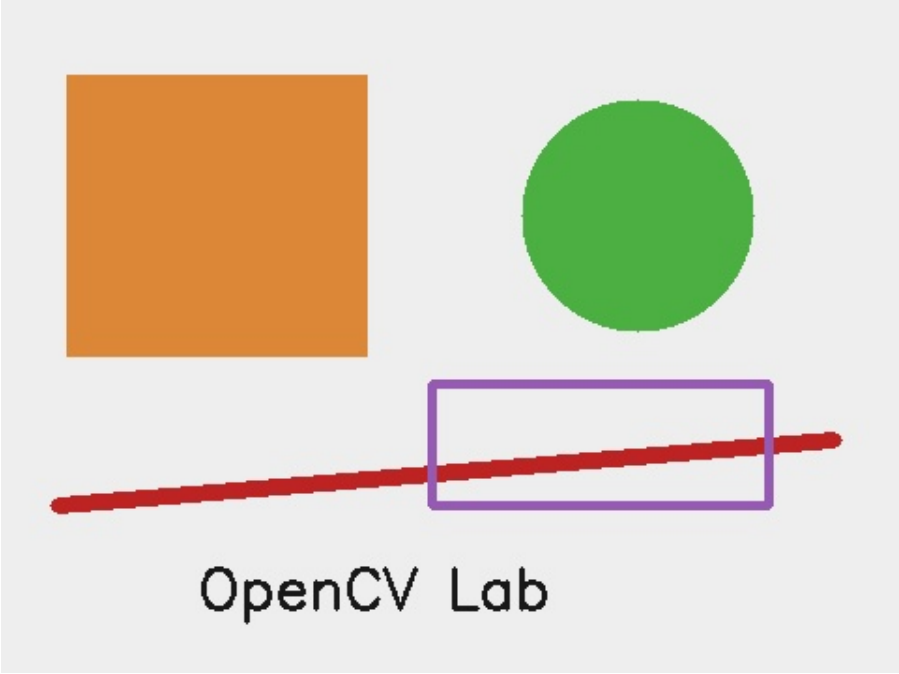
```
import cv2

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sx = cv2.convertScaleAbs(cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3))
sy = cv2.convertScaleAbs(cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3))
result = cv2.addWeighted(sx, 0.5, sy, 0.5, 0)
cv2.imwrite("output_images/experiment16_sobel.png", result)
print("Saved output_images/experiment16_sobel.png")
```

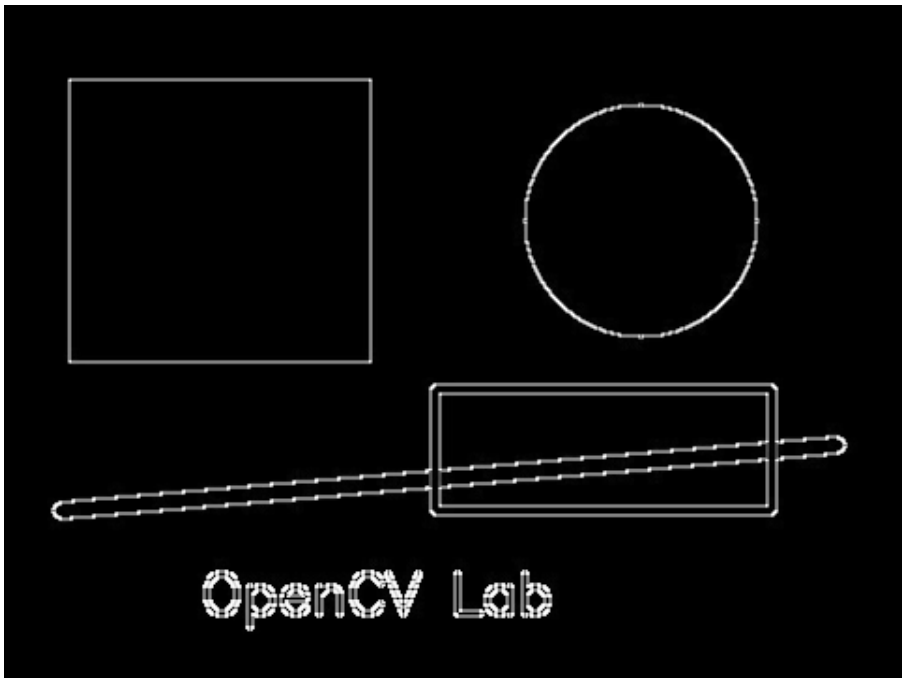
Input Image:

Output Image:

Input Image



Output Image


OpenCV Lab